

Assignment 2

due date: May 24, 2024

1. Suppose $\{X_n, n = 0, 1, 2, \dots\}$ is a two state Markov chain whose transition probability matrix P is given by

$$P = \begin{pmatrix} \alpha & 1 - \alpha \\ 1 - \beta & \beta \end{pmatrix}.$$

Let $Z_n = (X_{n-1} \ X_n)$ be a vector with four states $(0, 0), (0, 1), (1, 0), (1, 1)$. Is Z_n a Markov chain? If yes, determine its transition probability matrix.

2. A Markov chain with state space $\{0, 1, 2, 3\}$ has transition probability matrix

$$P = \begin{pmatrix} 1/2 & 1/4 & 0 & 1/4 \\ 0 & 1 & 0 & 0 \\ 0 & 1/4 & 1/2 & 1/4 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

- (a) Draw the transition diagram.
- (b) Suppose the initial probability distribution is uniform, i.e. $P(X_0 = i) = 1/4, i = 0, 1, 2, 3$. Determine $P(X_n = 1)$ and $P(X_n = 2)$.
3. Consider independent tosses of a fair die. Let X_n be the maximum of the numbers appearing in the first n throws, $n = 1, 2, \dots$
- (a) Verify that $\{X_n : n \geq 1\}$ is a Markov Chain. Describe the state space and the transition probability matrix of this homogeneous Markov chain.
- (b) Find out the classes of all states. Is there any state closed? Is there any state recurrent/transient? Is there any state positive recurrent? Write your reasons briefly.
4. Assume a Markov chain X_n has states $0, 1, 2, \dots$ and transition probabilities:

$$p_{i,0} = \frac{1}{2i^\alpha}, \quad p_{i,i+1} = 1 - \frac{1}{2i^\alpha}, \quad \text{for } i > 0; \text{ and } p_{01} = 1$$

where $\alpha > 0$ is a constant.

- (a) Is this Markov chain irreducible?
- (b) Assume $X_0 = 0$ and let T be the first return time to 0 (i.e. the first time after the initial time the chain is back at the origin). Determine for which $\alpha > 0$ is

$$1 - f_0 = P(\text{no return}) = \lim_{n \rightarrow \infty} P(T > n) = 0.$$

- (c) Depending on α , determine which classes are recurrent.

5. Let $\{X_n, n = 0, 1, 2, \dots\}$ be a simple random walk. Each step it has equal probability to walk to the left and to the right. The first passage time

$$\tau(m) = \min\{n \geq 0 : X_n = m\}$$

is the first time the process ever walks to position m . Suppose that $X_0 = 0$ and denote $G_m(z)$ as the probability generating function of $\tau(m)$ (i.e. $G_m(z) = \mathbb{E}z^{\tau(m)}$). Without loss of generality, assume that $m > 0$.

(a) Show that $G_m(z) = (G_1(z))^m$.

(b) Use the result of part (a), or otherwise, to find $G_m(z)$.